# [T]-Theory: A Universal Theory of Everything

One Equation. 61 Decades. Mind, Body, and Cosmos.

*Why:* a Green's function is not just a technical solution; it is the most direct account of how a disturbance becomes an effect. [T]-Theory uses that common response law to connect field theory, compactification, and the cosmological questions of dark energy and dark matter.

**Proved (Lean 4, 0 sorries):**

- SHO = Green's function (derived, not postulated as a primitive)
- 11D manifold = minimum geometry for a conscious vertebrate organism
- Scale invariance $10^{-35}$m $\to$ $10^{26}$m (61 orders of magnitude)
- Theory is a proposed fixed-point attractor of its own subject matter (see the [T]-Theory phenomena paper)

**Numerical hits (Planck 2018):**

- $\Omega_\Lambda = 7/11$ (**93.2%**) — dark energy as compactified vacuum
- $\Omega_c = 3/11$ (**97.1%**) — dark matter as spatial vacuum; gravity only; no direct detection possible
- $w = -1$ exact — live knife-edge test: DESI DR1 & Euclid

*[T]-Theory accounts for* ***95.5%*** *of the total mass-energy of the universe via a zero-parameter geometric derivation.*

## I · The Master Field Equation

The Universal Somatic Field is governed by a single structural equation — the Green's function of a tensor field derived from M-theory compactification.

**Stationary form (Helmholtz):**

$$(\nabla^2 + k^2)\, G(x, x') = -\delta^3(x - x')$$

**Time-dependent form (d'Alembertian):**

$$\left(\frac{1}{v_s^2}\partial_t^2 - \nabla^2 + k^2\right) \Phi_{\mu\nu}(x,t) = -J_{\mu\nu}(x,t)$$

**Retarded propagator** — causal, $\tau = t - t' > 0$:

$$G_R(r,\tau) = \frac{v_s}{4\pi r}\, e^{-kv_s\tau}\, \delta(\tau - r/v_s)\, \theta(\tau)$$

**Somatic Memory Kernel:**

$$K(\tau) = K_0\, e^{-\tau/\tau_m}\, \theta(\tau), \quad \tau_m = \frac{1}{kv_s}$$

**General solution** (field = integral of all past sources):

$$\Phi(x,t) = \Phi^{(0)}(x,t) + \int_{-\infty}^{t} dt' \int d^3x'\; G_R\, J$$

The **effective mass** $k$ sets the spatial correlation length $\ell = 1/k$. As $k \to 0$ (near consciousness threshold $T_c$), $\ell \to \infty$: global integration onset.

The **field temperature** $T_{\text{field}}$ controls how freely the field explores the attractor landscape. Therapeutic window: $T_{\text{field}} \in [T_{\min}, T_{\max}]$.

## II · Type-Theoretic Architecture (HoTT / Lean 4)

The soma-field has a rigorous type-theoretic formulation in Homotopy Type Theory, machine-verified in Lean 4 using Mathlib and Physlib.

**The Σ-type** — soma-field as a fiber bundle over the 20-scale base:

$$\text{SomaField} \;\equiv\; \sum_{(\sigma\,:\,\text{Scale}_{20})} \text{Substrate}(\sigma)$$

where $\text{Substrate}(\sigma)$ : Type is the physical substrate at scale $\sigma$. The base space is the 20-point scale hierarchy; each fiber is a field configuration.

**The Zoom Operator** — dependent type constructor between adjacent fibers:

$$\Lambda : (\sigma : \text{Scale}_{20}) \to \text{Substrate}(\sigma) \to \text{Substrate}(\sigma + 1)$$

**Causality as a proof argument** — the retarded propagator's full type:

$$G_R : (t\ t' : \text{Time}) \to (t' < t) \to \text{Space} \to \text{Space} \to \text{Field}$$

The inequality $t' < t$ is a *proof term* — the Lean 4 kernel enforces the arrow of time at compile time. A causal violation is a type error, not a runtime error.

**The full USF type:**

$$\text{USF} \;\equiv\; \sum_{(\sigma\,:\,\text{Scale}_{20})} (\text{Substrate}(\sigma) \times G_R(\sigma))$$

**BFSS cortex** — coordinates emerge from Hermitian